

# Network Types -- Guided Notes

Unit tp-2.9 | CompTIA Network+ N10-009 Obj. FC0-U71 2.9 | N10-008 FC0-U71 2.9

Name: \_\_\_\_\_ Date: \_\_\_\_\_ Period: \_\_\_\_\_

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Complete each question using the lesson reading. Write in the blank or answer in the space provided.

## Question 1

A LAN (Local Area Network) covers a \_\_\_\_\_ geographic area such as a single \_\_\_\_\_ or office. Devices on a LAN communicate at \_\_\_\_\_ speeds (typically 1 Gbps on modern wired networks). The key device connecting LAN devices is the \_\_\_\_\_.

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

## Question 2

A WAN (Wide Area Network) spans \_\_\_\_\_ geographic areas and connects multiple LANs. The Internet is the largest example of a \_\_\_\_\_. WANs use \_\_\_\_\_ provided by ISPs (leased lines, fiber, etc.) to link distant locations. WAN links typically have \_\_\_\_\_ (higher/lower) latency than LAN.

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

## Question 3

A WLAN (Wireless LAN) uses \_\_\_\_\_ signals instead of physical cables to connect devices. The IEEE standard for Wi-Fi is \_\_\_\_\_. A WLAN is still a \_\_\_\_\_ -- it just uses wireless transmission within that local area. A \_\_\_\_\_ provides the wireless signal in a WLAN.

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

## Question 4

A MAN (Metropolitan Area Network) is larger than a \_\_\_\_\_ but smaller than a \_\_\_\_\_. It typically covers a \_\_\_\_\_, campus, or city district. Examples include a city-wide cable TV network or a \_\_\_\_\_ network connecting multiple university buildings.

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

## Question 5

A PAN (Personal Area Network) covers a very \_\_\_\_\_ range (typically \_\_\_\_\_ meters) and connects personal devices near a single individual. \_\_\_\_\_ and \_\_\_\_\_ are the most common PAN technologies. An example is a laptop connected to wireless earbuds.

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

## Question 6

A VPN (Virtual Private Network) creates an \_\_\_\_\_ tunnel over a public network (like the Internet), allowing remote users to access a \_\_\_\_\_ network as if they were physically present on it. Data inside the VPN tunnel is \_\_\_\_\_, protecting it from interception on the public network.

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#### Question 7

An intranet is a \_\_\_\_\_ network accessible only to authorized internal users within an organization. An extranet extends access to \_\_\_\_\_ (partners, vendors) with appropriate permissions. Neither is the same as the \_\_\_\_\_, which is a public network accessible to anyone.

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#### Question 8 -- Real World Application

REAL WORLD: A company has offices in Austin, TX and Boston, MA. Each office has its own LAN. Developers in both offices need to access the same internal code repository server located in Austin. Describe the network types involved and explain how a VPN could be used to give the Boston team secure access to the Austin server across the Internet.

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*Download the answer key from the class server:*

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wget http://[SERVER]/resources/network-engineer/tp-2-9-network-types/tp-2-9-network-types-answer-key.pdf
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